

Assignment 3: Encoder vs. Decoder Language Models for Offensive Language Detection

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1 Introduction

This assignment aims to classify tweets as offensive or non-offensive using two baseline models: an encoder-only model (RoBERTa-base) and a decoder-only model (Qwen3-0.6B). Both models were fine-tuned to improve performance, and are also evaluated zero-shot to isolate the benefit of fine-tuning.

Encoder-only and decoder-only transformers differ in three key respects. Encoders use bidirectional attention, so each token attends to the entire input at once, and are pretrained with a masked-language-modeling objective; they are typically used for classification and other tasks requiring a single fixed representation of the input. Decoders use causal (unidirectional) attention, so each token only attends to preceding ones, and are pretrained purely for next-token prediction over much larger and more general corpora; they are typically adapted to a task through instruction fine-tuning.

The dataset used was TweetEval-Offensive. Each row contains an ID, text, and a label classifying the tweet as offensive (1) or non-offensive (0). The dataset contains 11,916 training, 1,324 validation, and 860 test tweets. It is hypothesized that the fine-tuned encoder will slightly outperform the fine-tuned decoder, since bidirectional attention and a dedicated classification head are traditionally better suited to short-text classification than an instruction-following decoder trained via LoRA.

As bonus tasks, both fine-tuned models were trained on shrinking fractions of the training set to determine which architecture is more sample-efficient. Furthermore, a token-attribution method is applied to both fine-tuned models to identify which words are most influential for each prediction.

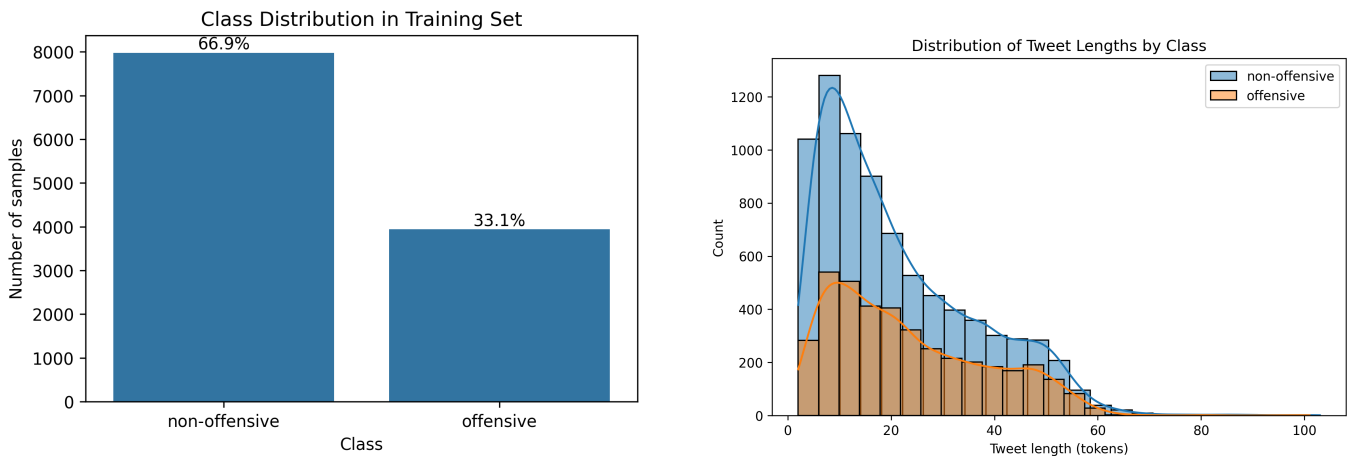
2 Methods and Results

2.1 Data

Class distributions were imbalanced in the training set, with the offensive class accounting for approximately 33% of examples (Figure 1). Both models used a class-weighted loss during fine-tuning to counteract this imbalance. Tweet length was right-skewed for both classes (Figure ??), with offensive tweets slightly longer on average (mean 23.26 tokens, median 20.00, $n = 3941$) than non-offensive tweets (mean 21.41, median 17.00, $n = 7975$); standard deviations were comparable (14.98 vs. 15.11).

Two separate preprocessing pipelines were used: the encoder pipeline tokenizes tweets directly with the RoBERTa tokenizer (max length 128, padding/truncation applied), while the decoder pipeline wraps each tweet in a single user-role instruction prompt and applies Qwen3's chat template before tokenization, framing classification as instruction-following rather than direct label prediction.

Figure 1: Class distribution and tweet length in the training set.



2.2 Models

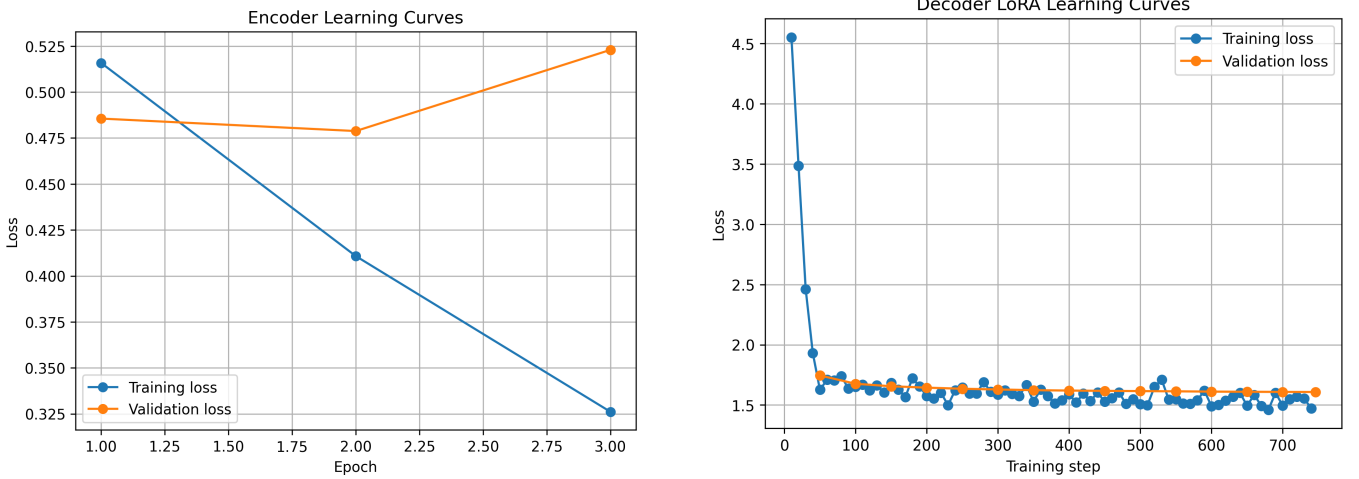
Four conditions were evaluated: an encoder zero-shot baseline, a fine-tuned encoder, a decoder zero-shot baseline, and a fine-tuned decoder. The encoder was fine-tuned for 3 epochs with early stopping (patience 2), while the decoder was fine-tuned for 2 epochs via LoRA.

- **Model 1: RoBERTa Zero-Shot:** RoBERTa-base with a randomly initialized linear classification head attached to the [CLS] token representation, evaluated without any task-specific training.
- **Model 2: RoBERTa Fine-Tuned:** The same architecture, fully fine-tuned end-to-end on the training set with a class-weighted cross-entropy loss, AdamW optimizer, and early stopping on macro F1. Trainable parameters: 124,647,170 trainable / 124,647,170 total (100%).
- **Model 3: Qwen3-0.6B Zero-Shot:** The pretrained instruction model prompted directly with an instruction describing the classification task, with the predicted label extracted from the generated text. Both Qwen3 conditions use the same prompt template: a user message reading *"Classify the following tweet as exactly one label: Offensive or Non-Offensive. Tweet: {text}. Answer with only the label."*, with the predicted label parsed from the generated completion.
- **Model 4: Qwen3-0.6B LoRA Fine-Tuned:** The same base model fine-tuned with LoRA (rank 16, alpha 32, dropout 0.1), targeting all attention projection matrices (`q_proj`, `k_proj`, `v_proj`, `o_proj`) and all MLP projection matrices (`gate_proj`, `up_proj`, `down_proj`) in every transformer layer, framing classification as instruction-following. Trainable parameters: 10,092,544 trainable / 606,142,464 total (1.6650%).

2.3 Training

The encoder’s training loss decreased steadily across epochs while validation loss increased after the second epoch. The decoder’s training and validation loss decreased steadily across epochs but validation loss remained over 1.5.

Figure 2: Training and validation loss curves for the encoder and decoder.



2.4 Evaluation

Model performance was evaluated using accuracy, precision, recall, and macro-averaged F1-score on the validation set (Table 1). Macro F1 was used as the primary selection criterion since it weights both classes equally despite the training-set imbalance.

Table 1: Validation set performance across all four conditions.

Condition	Accuracy	Precision	Recall	Macro F1
Encoder zero-shot	0.65	0.33	0.50	0.40
Encoder fine-tuned	0.80	0.78	0.79	0.78
Decoder zero-shot	0.35	0.56	0.50	0.27
Decoder fine-tuned (LoRA)	0.78	0.76	0.79	0.77

Both zero-shot models almost always predicted a single class, showing no task-specific skill, while both fine-tuned models improved substantially, with the encoder slightly ahead in accuracy and macro F1. Similar results held on the test set (Table 2; Figure 3).

Table 2: Test set performance for both fine-tuned models.

Condition	Accuracy	Precision	Recall	Macro F1
Encoder fine-tuned	0.83	0.79	0.81	0.80
Decoder fine-tuned (LoRA)	0.82	0.78	0.81	0.79

Figure 3: Confusion matrices for the fine-tuned encoder and decoder on the test set.

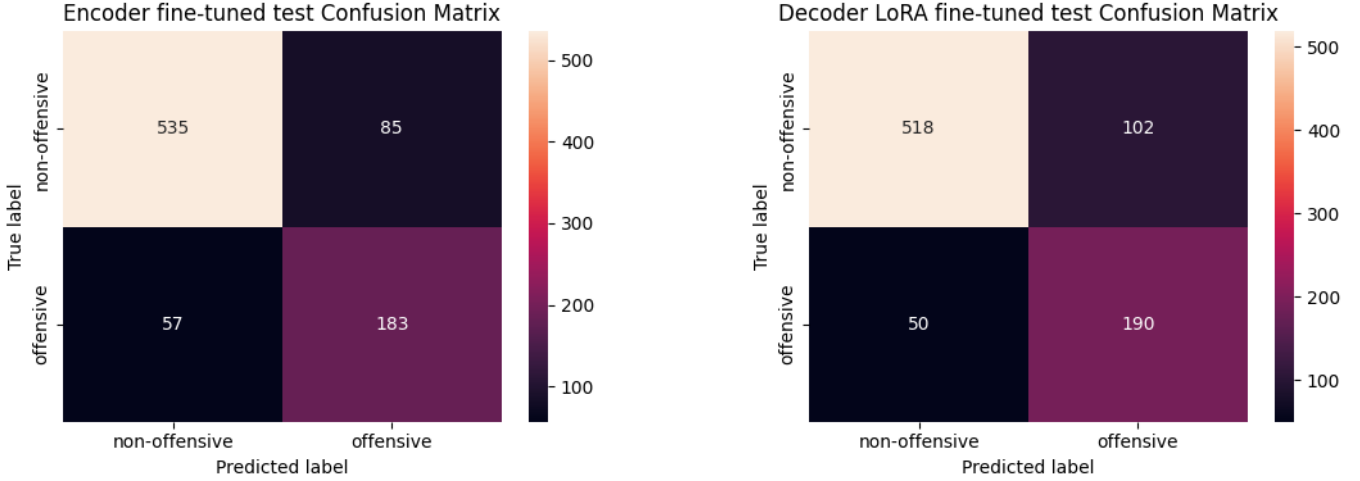


Table 3 shows two disagreements, both correctly resolved by the decoder but missed by the encoder: an informally-worded non-offensive tweet the encoder over-flagged as offensive, and a seemingly complimentary but objectifying tweet the encoder under-flagged as non-offensive. This suggests the decoder may be somewhat more robust to informal language and implicit tone, despite the encoder’s slightly higher aggregate test-set score.

Table 3: Representative disagreements between the fine-tuned encoder and decoder on the test set.

Tweet	True Label	Encoder Pred.	Decoder Pred.
“if he is im getting turnt with the holo keef regardless”	non-offensive	offensive	non-offensive
“Awwww she is so stinking cute! How old is she now?”	offensive	non-offensive	offensive

3 Conclusion

The hypothesis that the fine-tuned encoder would outperform the fine-tuned decoder was supported, confirming that bidirectional attention and a dedicated classification head give a modest but consistent edge for short-text classification tasks.

In terms of practical trade-offs, the encoder was cheaper to fine-tune given its much smaller size, while the decoder relied on LoRA to update only a small fraction of its parameters, making its fine-tuning far more memory-efficient despite the larger base model. At inference time, the encoder is a single forward pass, while the decoder must generate its answer token by token, making it slower to deploy at scale.

Limitations include the lack of a hyperparameter search, a decoder zero-shot baseline sensitive to unexplored prompt phrasing, and results from a single random seed; future work could address these by running a hyperparameter search, systematically varying the zero-shot prompt, or averaging results across multiple seeds.

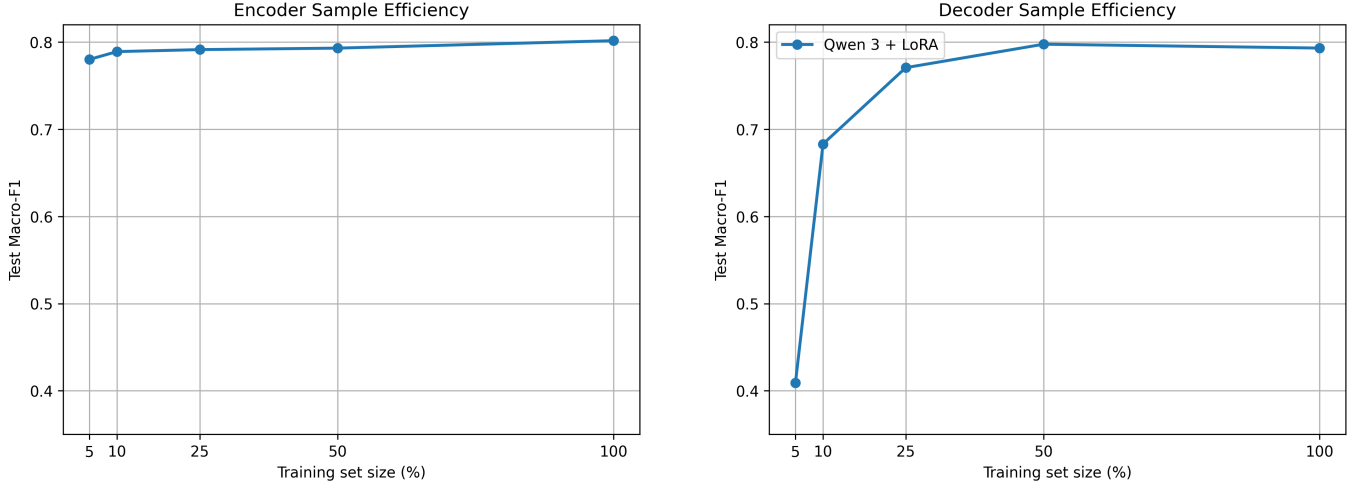
Bonus Task 1: Data Efficiency

Both fine-tuned models were retrained on shrinking fractions of the training set (50%, 25%, 10%, 5%), with fixed validation and test sets, to assess sample efficiency (Figure 4). The encoder was stable across training set sizes, reaching a macro F1 of 0.78 with just 5% of the data, only 0.02 below its full-data result of 0.80. The decoder collapsed at 5% (macro F1 0.41, barely above its 0.27 zero-shot baseline), but recovered quickly, matching the encoder from 50% onward (0.80 vs. 0.79). This suggests the decoder’s broad pretraining does not translate into sample efficiency under LoRA with very few examples, while the encoder’s classification head adapts to limited data more readily.

Bonus Task 2: Attention and Attribution Analysis

Integrated Gradients were applied to both fine-tuned models on a correctly classified and a misclassified tweet (Figure 4). For the correct example, the encoder attributed its decision to explicit insult words, while the decoder’s attribution was diffuse across the whole prompt, including instruction boilerplate. For the misclassified tweet, the encoder attributed strongly to the neutral word “Ratings” rather than the dismissive phrasing, failing to detect sarcasm; the decoder showed no clear localized signal. Both architectures rely heavily on explicit vocabulary and struggle with tone-driven offense lacking a trigger word. The encoder’s attributions are sparse and interpretable; the decoder’s are diffuse, showing that its attribution target (first token of the predicted label) is a proxy rather than a direct classification logit.

Figure 4: Bonus task results.



Tweet [0]: #ibelieveblaseyford is liar she is fat ugly libreal #snowflake she sold her herself to get some cash !! From dems and Iran ! Why she

Legend: ■ Negative ■ Neutral ■ Positive

True Label	Predicted Label	Attribution Label	Attribution Score	Word Importance
offensive	offensive (1.00)	offensive	2.48	#s # ibel iev e bl ase y ford is liar she is fat ugly lib real # s now flake she sold her herself to get some cash !! From dem s and Iran ! Why she spoke after # John K erry Iran Me eting ? #/s

Tweet [46]: #EmmyAwards2018 - Ratings tank as expected. Who can possibly stand to watch so many liberals in one place at one time.

Legend: ■ Negative ■ Neutral ■ Positive

True Label	Predicted Label	Attribution Label	Attribution Score	Word Importance
offensive	non-offensive (1.00)	non-offensive	1.66	#s # Em my A wards 2018 - Ratings tank as expected . Who can possibly stand to watch so many liberals in one place at one time . #/s

(a) Encoder token attributions.

Tweet [0]: #ibelieveblaseyford is liar she is fat ugly libreal #snowflake she sold her herself to get some cash !! From dems and Iran ! Why she

Legend: ■ Negative ■ Neutral ■ Positive

True Label	Predicted Label	Attribution Label	Attribution Score	Word Importance
offensive	offensive (1.00)	offensive	3.09	# im_start user Class ify the following tweet as exactly one label : Offensive or Non - Off ensive . Tweet : # ibel iev e bl ase y ford is liar she is fat ugly lib real # snow flake she sold her herself to get some cash !! From dem s and Iran ! Why she spoke after # John K erry Iran Meeting ? Answer with only the label . # im_end # im_start assistant #think #/think

Tweet [46]: #EmmyAwards2018 - Ratings tank as expected. Who can possibly stand to watch so many liberals in one place at one time.

Legend: ■ Negative ■ Neutral ■ Positive

True Label	Predicted Label	Attribution Label	Attribution Score	Word Importance
offensive	non-offensive (1.00)	non-offensive	3.51	# im_start user Class ify the following tweet as exactly one label : Offensive or Non - Off ensive . Tweet : # Em my A wards 2018 - Ratings tank as expected . Who can possibly stand to watch so many liberals in one place at one time . Answer with only the label . # im_end # im_start assistant #think #/think

(b) Decoder token attributions.